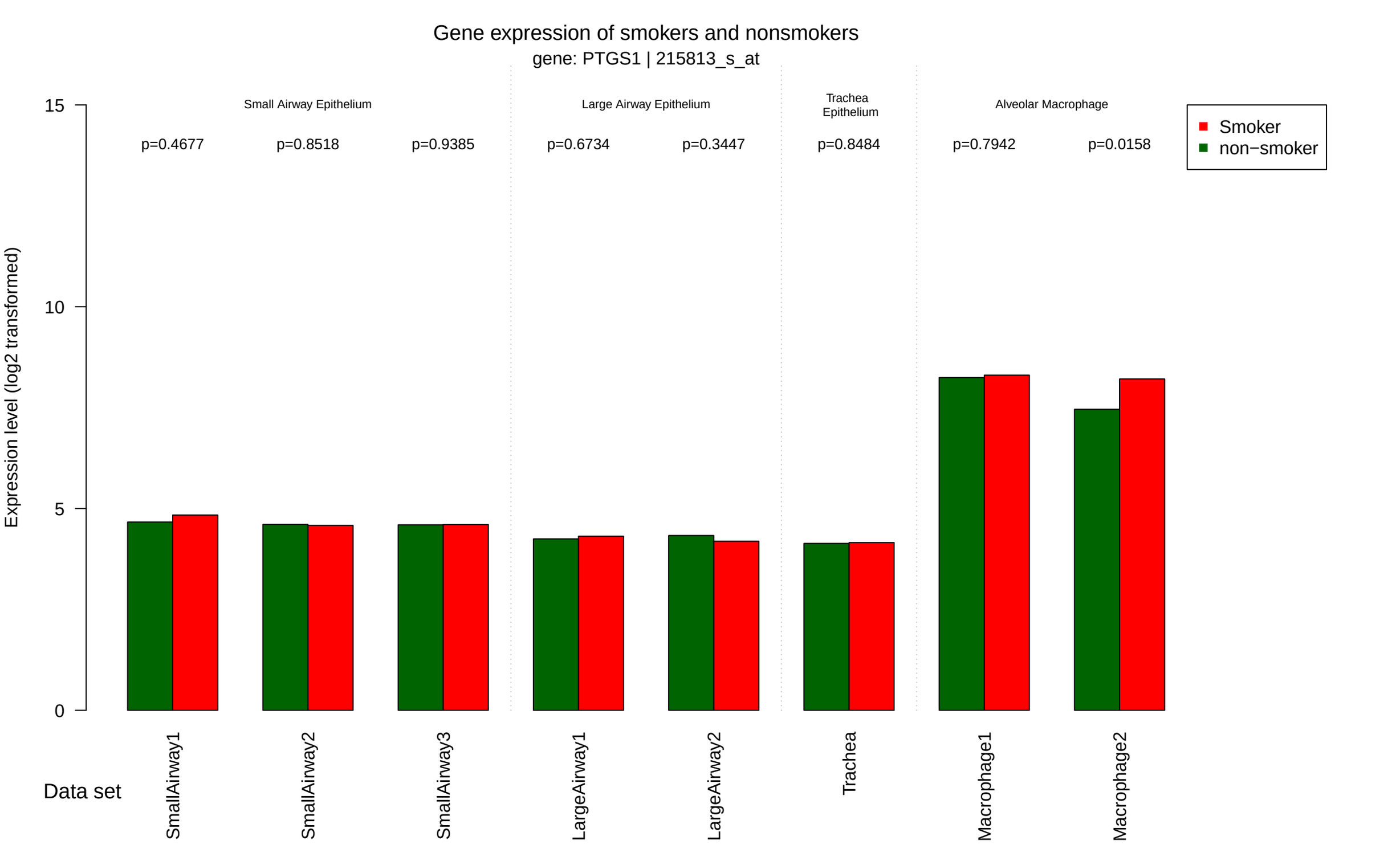
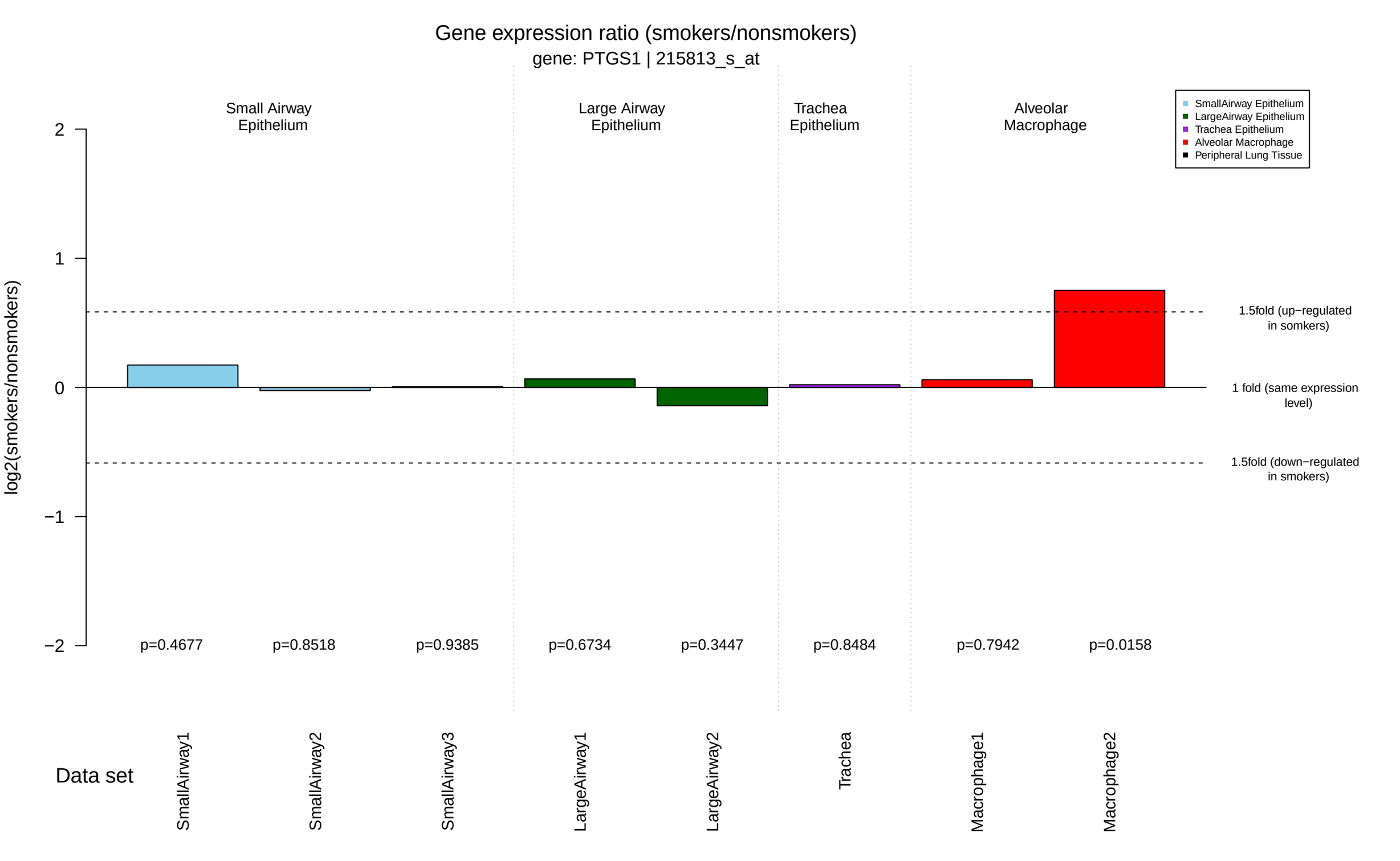
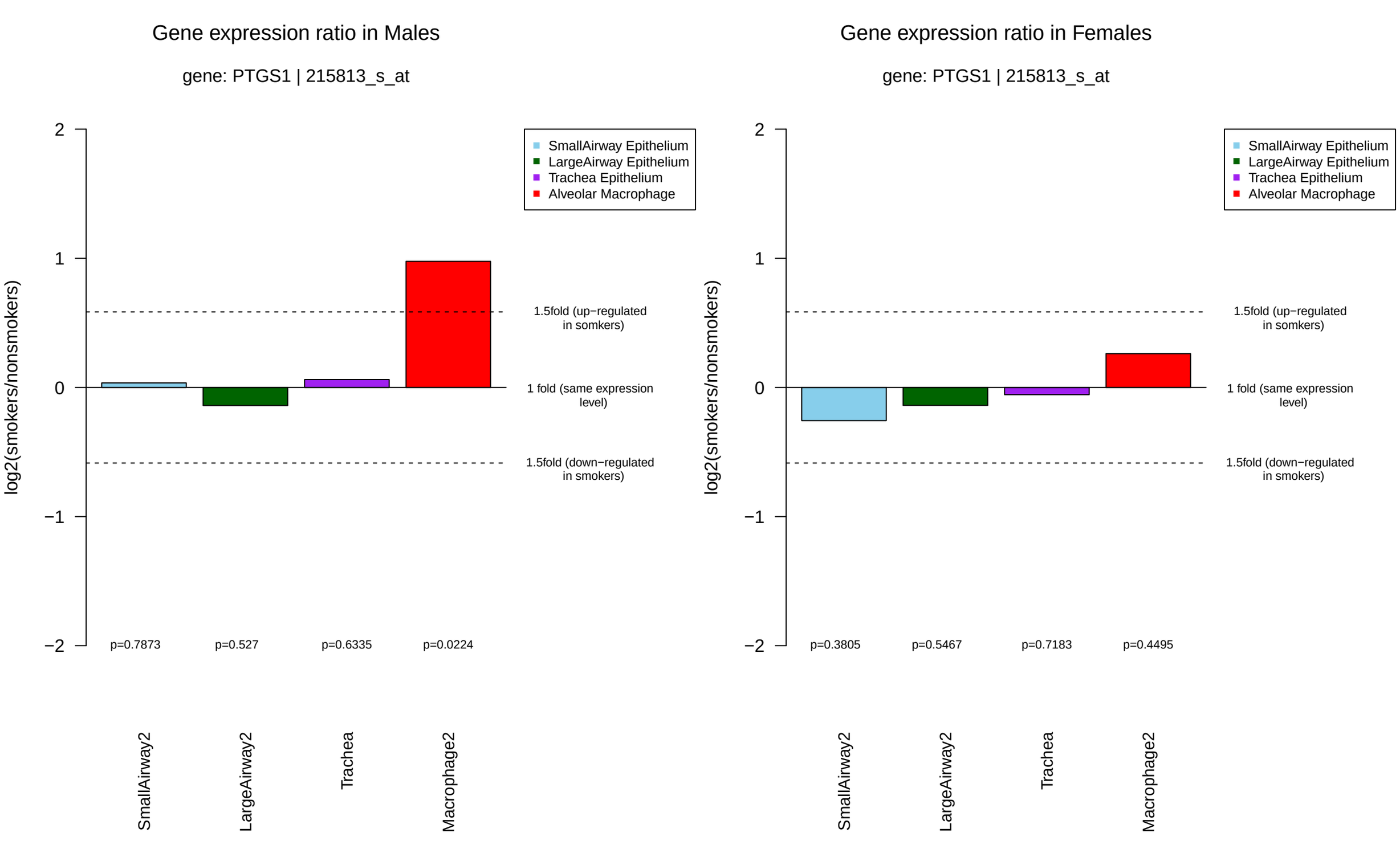
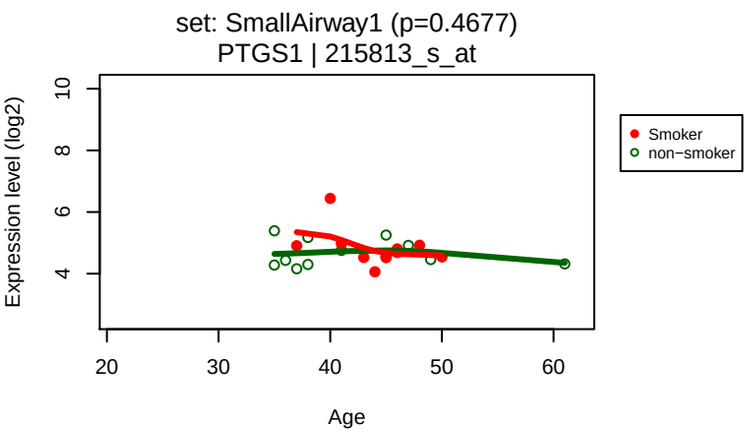
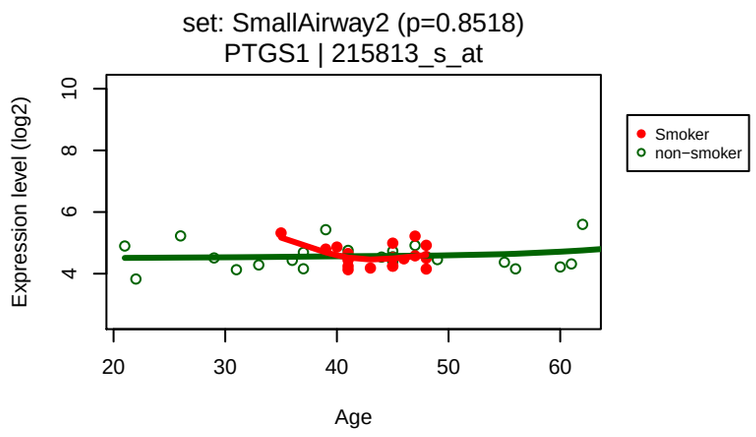
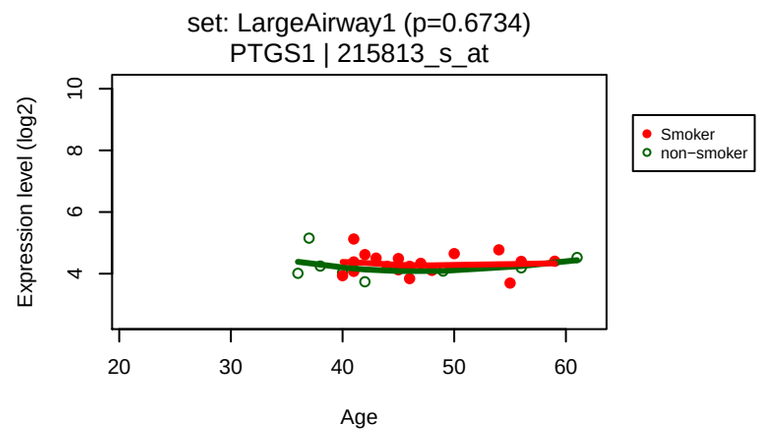

Gene expression of smokers and nonsmokers
gene: PTGS1 | 215813_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4677
p=0.8518
p=0.9385
p=0.6734
p=0.3447
p=0.8484
p=0.7942
p=0.0158
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

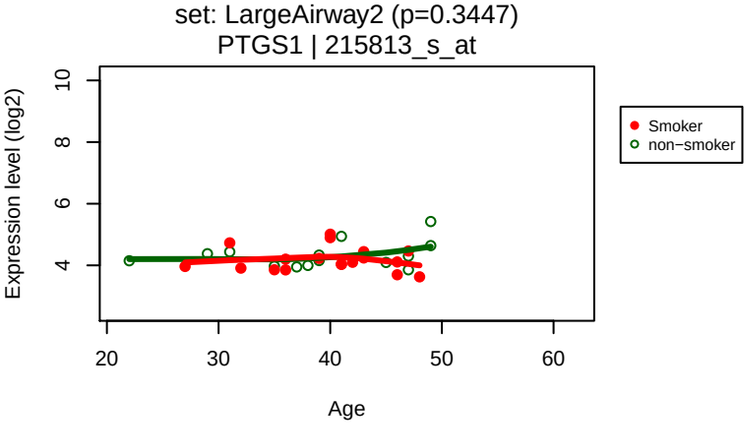

Gene expression ratio (smokers/nonsmokers)
gene: PTGS1 | 215813_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4677
p=0.8518
p=0.9385
p=0.6734
p=0.3447
p=0.8484
p=0.7942
p=0.0158
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

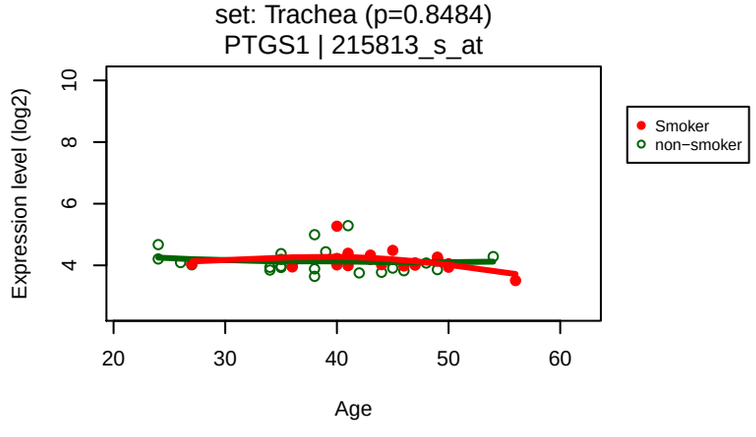

Gene expression ratio in Males
gene: PTGS1 | 215813_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7873
p=0.527
p=0.6335
p=0.0224
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTGS1 | 215813_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3805
p=0.5467
p=0.7183
p=0.4495
SmallAirway2
LargeAirway2
Trachea
Macrophage2

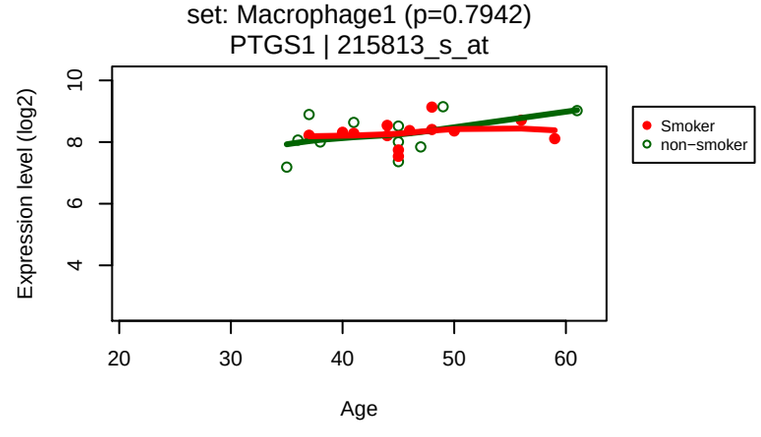

set: SmallAirway1 (p=0.4677)
PTGS1 | 215813_s_at
Expression level (log2)
Age
Smoker
non−smoker

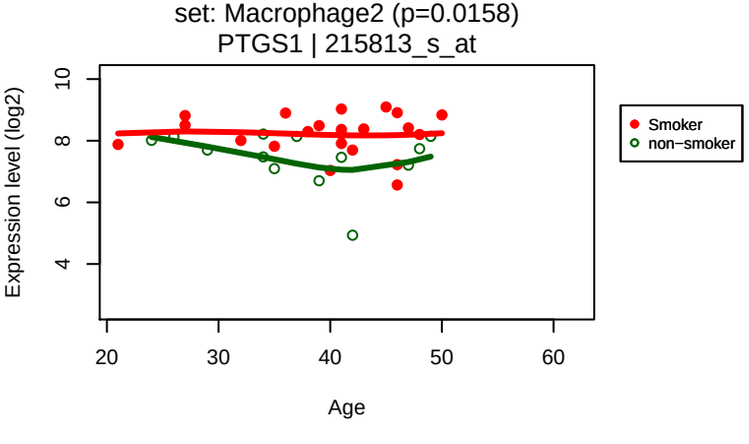

set: SmallAirway2 (p=0.8518)
PTGS1 | 215813_s_at
Expression level (log2)
Age
Smoker
non−smoker

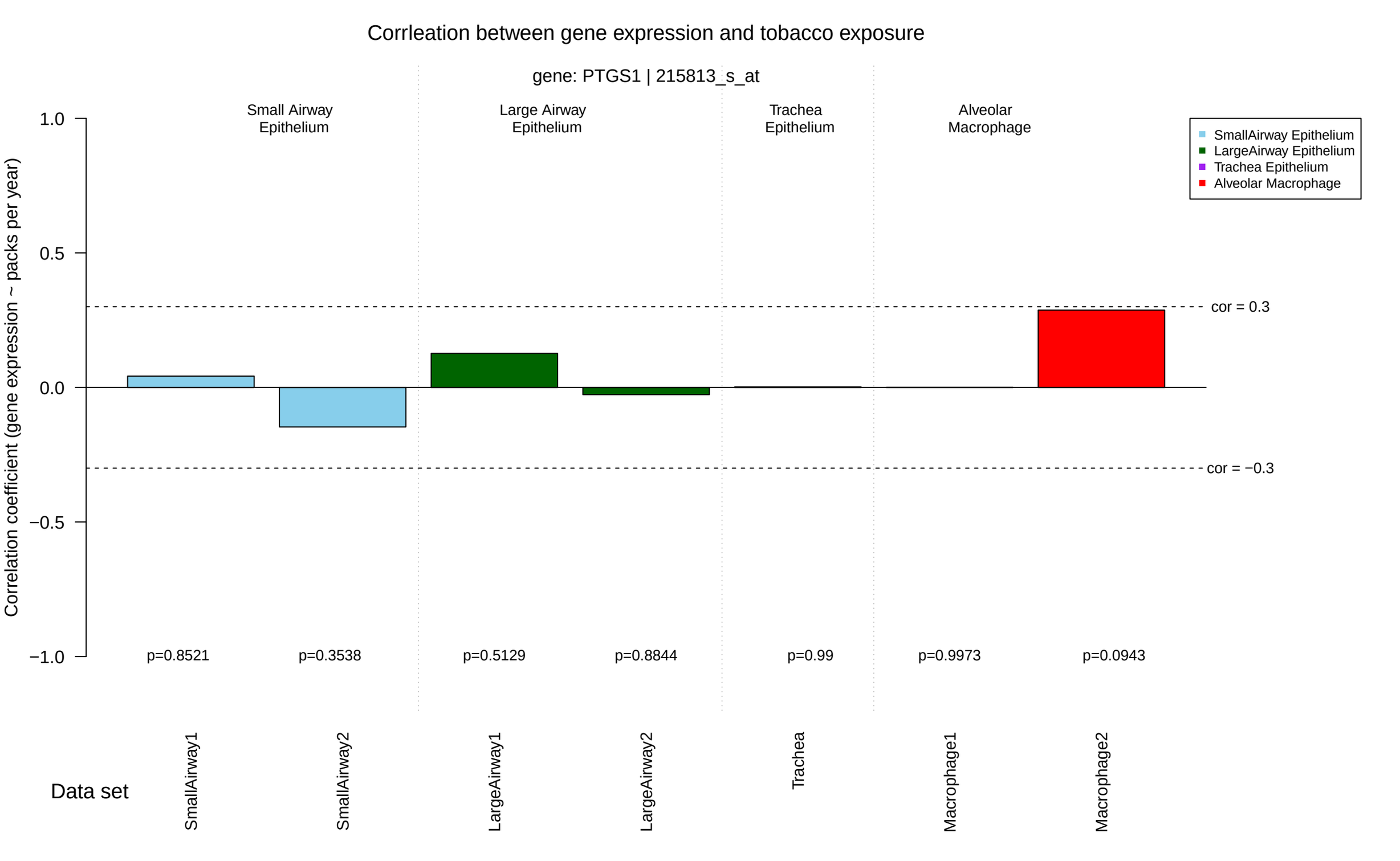

set: LargeAirway1 (p=0.6734)
PTGS1 | 215813_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3447)
PTGS1 | 215813_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8484)
PTGS1 | 215813_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7942)
PTGS1 | 215813_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0158)
PTGS1 | 215813_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTGS1 | 215813_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8521
p=0.3538
p=0.5129
p=0.8844
p=0.99
p=0.9973
p=0.0943
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2